

MANAR: A Pre-Flight Software Prototype, Route Optimization Heuristic, and Multi-Spectral Vision Benchmark for Aerial Search-and-Rescue

Two-Page Technical Brief and Architecture Overview

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Abstract—This brief presents MANAR (Multi-sensor Autonomous Navigation and Aerial Rescue), a pre-flight software prototype and supervised-autonomy framework for aerial Search-and-Rescue (SAR) robotics. Evaluated in-silico and on recorded video datasets prior to physical vehicle deployment, MANAR establishes a deterministic C++ control core, double-precision Haversine geodetics, an $O(n^2)$ greedy nearest-neighbor route optimizer with formal operator override privilege, sequential lawnmower sweeps, and an embedded ONNX Runtime visual detection pipeline. We present a 500-scenario route optimization evaluation against 2-opt, an empirical multi-spectral 26-video SAR benchmark ($> 4\,200$ annotated frames) comparing YOLO11n against D-FINE-N, an analytical telemetry power model, and a systematic software verification matrix.

Index Terms—Search-and-Rescue UAVs, Supervised Autonomy, Deterministic Control, Greedy Route Optimization, Haversine Geodetics.

I. INTRODUCTION AND SYSTEM MATURITY TIERS

Search-and-Rescue (SAR) operations in remote wilderness and desert environments demand rapid aerial coverage under severe time constraints. Fully manual teleoperation overwhelms pilot cognitive capacity during multi-sensor tracking, while unconstrained autonomous systems introduce severe failure risks. MANAR resolves this trade-off through a **supervised-autonomy model** where deterministic flight logic and advisory perception operate under continuous human supervisory authority.

To establish rigorous engineering grounding, MANAR categorizes all capabilities across three distinct maturity tiers (Table I):

- 1) **Implemented Functional Prototype:** Fully compiled C++ logic, double-precision Haversine navigation, $O(n^2)$ greedy route optimization with operator override (Y/N), lawnmower sweeps, battery failsafes, and structured event logging.
- 2) **Selected Closed Architecture:** Authoritative state ownership in C++, React/TypeScript interface via WebSocket + JSON, and on-demand Return-to-Launch (RTL) navigation semantics.
- 3) **Planned Future Work:** PyTorch perception models, Mamba temporal fusion, PX4 SITL/HITL simulation, and field validation.

TABLE I
MANAR CAPABILITY MATURITY STATUS

Subsystem	Component / Function	Maturity Tier
Control Core	C++ Flight Logic (<code>control.exe</code>)	✓ Implemented
	Double-Precision Haversine	✓ Implemented
	Route Planner	✓ Implemented
	Search Logic	✓ Implemented
	Safety System	✓ Implemented
Visual CV	YOLO11n + ORT C++ (29.2 ms)	✓ Implemented
CV Baseline	D-FINE-N 26-Video Benchmark	✓ Implemented
Operator GUI	React / TypeScript Web UI	▲ Selected
IPC Protocol	WebSocket + JSON Transport	▲ Selected
State Store	Authoritative C++ Core Sync	▲ Selected
Candidate Track	Kalman / ByteTrack MOT	○ Planned
Fusion Engine	Mamba State Space Model	○ Planned
Autopilot	PX4 SITL / HITL Simulation	○ Planned
Airframe	Physical Flight Deployment	○ Planned

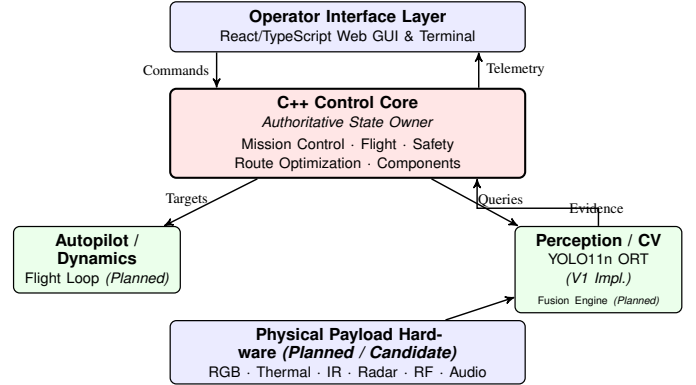


Fig. 1. MANAR system architecture. The C++ Control Core is the authoritative state owner governing navigation, safety, and routing.

II. IMPLEMENTED DETERMINISTIC C++ CORE [TIER 1]

The functional prototype (`core/`) executes deterministic navigation, route optimization, and safety monitoring:

- 1) **Haversine Geodetics:** Exact surface distance calculation over spherical coordinates (ϕ, λ) :

$$a = \sin^2 \left(\frac{\Delta\phi}{2} \right) + \cos \phi_1 \cos \phi_2 \sin^2 \left(\frac{\Delta\lambda}{2} \right) \quad (1)$$

$$d = 2R_E \operatorname{atan2} \left(\sqrt{a}, \sqrt{1-a} \right) \quad (2)$$

where $R_E = 6\,371\,000.0$ m.

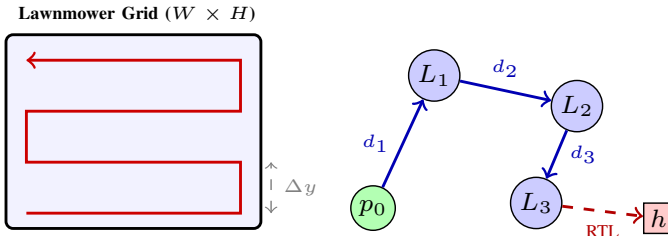


Fig. 2. Spatial geometry: Lawnmower sweep pattern (left) and sequential greedy route progression with Return-to-Launch final leg (right).

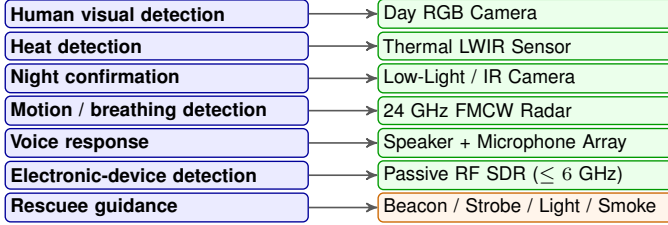


Fig. 3. Operational SAR capability to payload sensor and signaling subsystem mapping.

- 2) **Greedy Route Optimizer** ($O(n^2)$): For n search locations, the planner iteratively selects the nearest unvisited target:

$$p_{k+1} = \arg \min_{p \in U_k} d(p_k, p) \quad (3)$$

$$N_{\text{eval}} = \frac{n(n+1)}{2} = O(n^2) \quad (4)$$

For $n = 8$, this evaluates 36 candidate distances ($8 + 7 + \dots + 1 = 36$) vs. $8! = 40320$ permutations. Operators retain an explicit override toggle (Y/N) to preserve manual tactical ordering.

- 3) **Lawnmower Sector Sweeps**: Parameterized sweeps ($W \times H$, row spacing Δy) use exact local scaling constants:

$$M_{\text{lat}} = 111\,320.0, \quad M_{\text{lon}}(\phi_c) = 111\,320.0 \cos(\phi_c) \quad (5)$$

- 4) **Battery Failsafe Hierarchy**: Continuous monitoring of simulated battery B_k : Warning (30%), Recommend RTL (25%), Emergency RTL (15%), and Emergency Land (5%).

III. MULTISENSOR PAYLOAD & PERCEPTION [TIER 1 & 3]

MANAR incorporates eleven controllable payload categories for victim search, identification, and signaling (Fig. 3):

- **Visual Candidate Detection (YOLO11n)**: Evaluated across 26 multi-spectral SAR video datasets (> 4200 frames). Operates at 29.2 ± 2.4 ms average latency (34.2 ± 2.8 FPS on CPU, $1.73 \times$ faster than 50.5 ± 4.1 ms D-FINE-N baseline) with 100% clean negative rejection (0 FP) and superior concealed-target sensitivity (44 vs. 0 detections).
- **Diurnal Optical Switching**: Sensor modes transition dynamically with hysteresis based on ambient light: > 150 lx (RGB+LWIR), $50\text{--}150$ lx (RGB+IR+LWIR), and < 50 lx (IR+LWIR).
- **Acoustic Call-and-Listen Cycle [Tier 3]**: 35.4 s sequence (8.4 s SOS buzzer, 12.0 s Arabic/English voice prompt, 15.0 s silent listening with motor-RPM harmonic cancellation).

- **Passive RF Two-Level Attention [Tier 3]**: Level 1 CFAR threshold ($N_{\text{CFAR}+} \geq 3$) triggers Level 2 temporal trend tracking (ΔS_k) in Inspect mode. Civilian operations subject to telecommunications statutory clearance.
- **FMCW Radar Theoretical Ranging [Tier 3]**: 24 GHz radar ($B \approx 250$ MHz) provides $\Delta R \approx c/(2B) \approx 0.60$ m theoretical range resolution for breathing micro-motion.
- **Mamba SSM Alert Gate [Tier 3]**: Evaluates synchronized 10 Hz feature vectors $\mathbf{x}_t \in \mathbb{R}^D$ over 20–50 frames (2–5 s). Mamba gates alert escalation; human operator assigns final rescue status.
- **Rescue Guidance [Tier 3]**: 360° amber beacon, 2 Hz white strobe, spotlight, smoke marker, and 3 fixed heliograph mirrors ($40^\circ, 60^\circ, 75^\circ$).

IV. ADAPTIVE TELEMETRY POWER MODEL

Analytical incremental RF transmission power ΔP_i across frequencies f_i is formulated as:

$$\Delta P_i = (P_{\text{TX}} - P_{\text{RX}}) \cdot \left(\frac{8 \cdot B_i \cdot f_i}{R} \right) = \beta \cdot B_i \cdot f_i \quad (6)$$

where $P_{\text{TX}} = 4.8$ W, $P_{\text{RX}} = 0.36$ W, $R = 64$ kbps, and slope factor $\beta \approx 5.55 \times 10^{-4}$ W/(byte · Hz). Table II summarizes the 14-stream dual-regime transmission policy.

TABLE II
DUAL-REGIME TELEMETRY POLICY (14 STREAMS)

Stream Category	Payload (B_i)	Search	Proximity
Heartbeat / Position / Speed / Altitude	24–28 B	2.0 Hz	5.0 Hz
Heading	22 B	1.0 Hz	5.0 Hz
Battery Percentage	22 B	0.2 Hz	0.5 Hz
Flight / Mission / Component State	21–24 B	On Change	On Change
Raw Sensor Measurements	52 B	Local	5–10 Hz
Confidence Alerts / Failsafes	28–32 B	Event	Event
Diagnostics / Logs / Config	84 B	Pull	Pull

V. VERIFICATION AND SYSTEM ROADMAP

Software Prototype Verification: Software-in-the-loop integration testing confirmed deterministic C++ control flow, search plan locking, start rejection on empty plans, Return-to-Launch (RTL) preemption, and real-time ONNX Runtime visual candidate detection under Windows 11 / GCC.

System Roadmap:

- 1) *Interface Layer*: Complete React/TypeScript WebSocket GCS.
- 2) *Perception Tracking & Fusion*: Implement Kalman / ByteTrack tracking and Mamba temporal fusion.
- 3) *Autopilot Integration*: PX4 SITL simulation followed by HITL physical airframe flight testing.